ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

INFERENCE BY SAMPLING

SANJOY DASGUPTA, PROFESSOR

UC San Diego

COMPUTER SCIENCE & ENGINEERING
HALICIOĞLU DATA SCIENCE INSTITUTE

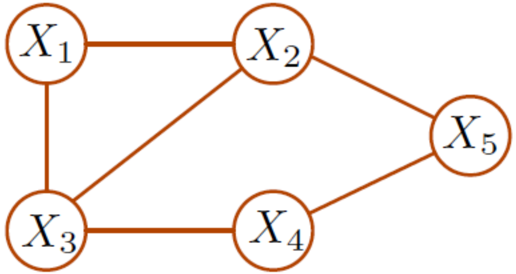
Recall: Markov Random Fields

Joint distribution over random variables $X_1, \ldots, X_n$ given by:

1. An undirected graph with nodes $X_1, \ldots, X_n$ and edges representing dependencies.
2. A distribution that factors over this graph:

$$P(X_1, \ldots, X_n) = \frac{1}{Z} \prod_C \psi_C(\{X_i : i \in C\})$$

where the product is over maximal cliques in the graph, and the **clique potentials** ψ_C are positive-valued functions.



Functional form of $P(X_1, X_2, X_3, X_4, X_5)$

$$\frac{1}{Z} \psi_{123}(X_1, X_2, X_3) \psi_{25}(X_2, X_5) \psi_{34}(X_3, X_4) \psi_{45}(X_4, X_5)$$

Three Types of Inference Queries

Suppose we have values for some of the variables, $X_E = x_E$ ("evidence").

1. **Conditional probability query:** What is the conditional distribution $\Pr(X_S | X_E = x_E)$ for some other variables S?
2. **Most probable explanation:** Find the highest-probability setting of the remaining variables.
3. **Maximum a posteriori:** Find the highest-probability setting for some other variables S.

Similar landscape to Bayes nets:

- All three types of query are NP-hard.
- Efficient exact inference for trees, or more generally, for bounded tree-width.
- Approximate inference using sampling, variational methods, belief propagation.

Inference Via Sampling

For a Markov random field $P(X_1, \dots, X_n)$, we have algorithms for:

- sampling $(X_1, \dots, X_n) \sim P$
- sampling $X_{\setminus \mathrm{E}} \sim P$, while “clamping" evidence $X_E = x_E$

Inference Via Sampling

For a Markov random field $P(X_1, \ldots, X_n)$, we have algorithms for:

- sampling $(X_1, \ldots, X_n) \sim P$
- sampling $X_{\setminus \mathrm{E}} \sim P$, while "clamping" evidence $X_E = x_E$

Can we use this for approximate inference?

1. Conditional probability queries $\Pr(X_i | X_E = x_E)$

2. Maximum a posteriori $\arg\max_{x_s} \Pr(X_S = x_S | X_E = x_E)$

Gibbs Sampler

To sample from a distribution P over $(x_1, \dots, x_n)$.

- Start with any x in the support
- Repeat:
  - Pick a feature $i \in \{1,2,\dots,n\}$
  - Resample x_i from $P\left(X_i = x_i | x_{\setminus \mathrm{i}}\right)$

E.g., if the X_i are $0-1$ valued then in each step:

- pick a feature i
- set $x_i = 1$ with probability

$$\frac{P\left(x_i = 1, x_{\setminus \mathrm{i}}\right)}{P\left(x_i = 0, x_{\setminus \mathrm{i}}\right) + P\left(x_i = 1, x_{\setminus \mathrm{i}}\right)}$$

Guaranteed to converge to the right distribution!

Energy-Based Models

- Any Markov random field

$$P(X_1, \ldots, X_n) = \frac{1}{Z} \prod_C \psi_C(\{X_i : i \in C\})$$

can be rewritten as

$$P(X) = \frac{1}{Z} e^{-U(X)}$$

- A distribution of this form is an **energy-based model**, and $U(x)$ is the **energy function**.

Gibbs sampling can be applied quite easily to any energy-based model with discrete variables.